Augmented Social Streetlamps

Citizen-centric Design of projections of third places for open XR and urban commons in inclusive smart cities

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ABSTRACT

Designing for young immigrants integrating in Malmo's sociocultural contexts finds in a citizen-centric approach to Urban Interaction Design and Mixed Reality an opportunity for inclusion, while tackling the digital divide. However, a lack of clarity on its definition as value and methodology within design research and HCI make it challenging to understand how this may impact the process. At the same time, the applications of augmented reality are vast and yet to be further understood.

This thesis explores how urban augmented media architecture may create socially sustainable and inclusive smart cities, putting the focus on citizens first. Interviews with an intersectional group of locals, a participatory approach to ideation and sketching, and some urban testing, together with visual demography, contextual observations, expertise from different global urbanisms, and the framework of Participant Journey Maps support the project. This results in a design research artefact to question the role of the networked digital layers, platforms, and services of the city as social media. Augmented Social Streetlamps not only provide a bottom-up proposal for the city, but also a case inviting a broader discussion around augmented public squares, playgrounds, and third spaces where citizens' knowledge can develop into urban commons.

KEYWORDS:

URBAN INTERACTION DESIGN, PARTICIPATORY DESIGN, SPATIAL AUGMENTED REALITY, SERVICE DESIGN, PUBLIC ARTS

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INTRODUCTION

When the design of a smart city goes beyond efficiency-driven digital and technological solutions, it can mediate new opportunities for interconnectedness through inclusivity, participation, and social sustainability. A citizen-centric approach to Urban Interaction Design explores this, utilising technologies and new media to foster communal engagement and democratic access to public spaces. The ubiquity of public smart technologies can provide new opportunities to engage in urban communities and landscapes, replacing expensive private devices which risk reinforcing socioeconomic disparities and exclusion. This thesis questions how public uses of Extended Reality may support this, particularly through visual projections (and projection mapping).

How can immersive installations and interactive media enhance public spaces to get people to connect offline? The participatory process involving ten citizens yields a co-design proposal for the city of Malmö, focusing on community building among young immigrant citizens, while providing accessible, equitable, and culturally inclusive digital services and urban experiences. It results in a lo-fi prototype that does not intend to bring definitive answers but to build a case of design research to discuss possible developments in the urban use of projections, for placemaking and social sustainability. Augmented squares and third spaces may conceal opportunities to create participation and social cohesion and to activate and reconnect different areas of the city through hybrid digiphysical spaces, where people can explore different shared expressions of citizenship and tackle city challenges together.

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Yuri (they/them) is part of the citizenship and communities of Malmö, but has Costa Rican–Italian origins, which influence their contributions. They try to acknowledge their position throughout the process, while bringing their lived experience as part of different minorities, and of the immigrant majority of the citizens of Malmö. The project and the designer also acknowledge the lands that hosted the process and its human and non-human citizens and communities.

The data gathered throughout the process has been gathered after informed consent and it has been handled in compliance with the European GDPR.

This thesis project uses AI tools to support grammar and syntax in its written presentation.

THEORETICAL BACKGROUND & STATE OF THE ART

“Information technology, tied as it is to our mental and cultural images of scientific representation and progress, is a tool not only for automation but also for legitimation"

(Dourish & Bell, 2011).

INCLUSIVE SMART CITIES

As technologies evolve, so do the conceptions of and expectations for North-Western cities and societies. On the one hand, our lifestyle has welcomed and accommodated the digital and virtual realms as an integral part of the lives of most, allowing us to widen our projections on what the urban futures could look like. On the other hand, the technocentric paradigm has had a foundational role in some understandings of the smart city – especially in Northern American academies – and consequentially in the way many cities have been developing (Mora et al., 2017), with a risk of reproducing urban inequities, even when nominations as “smart” or “inclusive” are used (Lee, 2020). This thesis finds context in the more holistic and human-centric European research around smart cities, which focuses on the generative power of citizens’ education, ideas, and creativity (Capello et al., 2011) "as the result of the balanced combination of human, social, cultural, economic, environmental, and technological aspects, which stand alongside one another" (Mora et al, 2017: 12).

While different interpretations of the term “smart cities” may impact their implementation, technologies remain a core element in their design, assisting localised groups of users through Information and Computer Technology (ICT), data, sensors, and media, and creating additional digital layers to the physical built environment. These may create urban living labs for systemic innovation (Schaffers & Turkama, 2012) and represent opportunities for connection: not only through digital infrastructures reaching out to the denizens to expand the services of the city but also uncovering the city as a multistakeholder platform that can directly benefit and empower their communities, providing people with interactive information, accessibility, sustainability, and safety to make them thrive.

“Smart” technologies cannot be the essence or the drive of a city’s development, where the inhabitants and the territory are the real cardinal points. “The qualifier smart, when applied to cities, is an empty signifier” (Mackinnon et al., 2022: 1) and the current Western approach to smart cities risks promoting a neo-liberal agenda charged with “pragmatic, instrumental and paternalistic discourses”, rather than prioritising the need for “an effective shift of power […] rooted in the right to the city, entitlements, community, participation, commons, and ideals beyond the market” (Cardullo & Kitchin, 2019: 12). People should come before technologies and business – inclusivity and participation over technological advancement.

After all, cities are the people that inhabit them and make them come to life: networks of citizens, before infrastructures and services, who ought to be at the centre of the negotiations for urban futures. The different institutional and professional actors involved in city-making – as urban planners, architects, engineers, and designers – are then jointly responsible for answering the needs, problems, and wishes of the citizens, requiring a multidisciplinary and participatory approach to design their surroundings, therefore their behaviours.

“Inclusion” may also be a deceiving adjective when smart cities still struggle to accommodate the needs of people with disabilities and older adults (Soares de Oliveira Neto & Kofugi, 2016; Lee, 2020) or peripheral and rural disadvantaged neighbourhoods (Kaharevic & Wihlborg, 2024). True inclusiveness should result in equal, just, and liveable smart cities (Lee, 2020). As ambitious as it may sound, a truly inclusive smart city is for all, and this means creating opportunities for participation and representation of intersecting diversity and pluriversal realities. This applies to places and technologies, as well as their urban digital and virtual layers. As technologies become capillary and unescapable, this also means that cities must “eliminate, not only physical but also digital, barriers”, while remembering that “citizens’ needs must be in the forefront” (Soares de Oliveira Neto, 2016).

URBAN INTERACTION DESIGN

Emerging design approaches are bringing user-centred design (Norman & Draper, 1986) and human-centred methods (Brown & Katz, 2009) to how cities are planned, applying them to urban services, experiences, and installations. Focusing on citizens as users – and involving them as co-creators, Urban Interaction Design (UIxD) aims at devising software layers and interactive services to enhance their experience and participation in the city, trying to design out the risk of social exclusion and putting people over profit (Brynskov et al., 2014; Pozzi & Bagnara, 2015). It proposes a relational approach to city-making, where designers, academics, policymakers, and citizens come together to tackle wicked urban problems, also through the technologies of the networked city, creating digiphysical interspaces (Winograd, 1997) for citizen participation. This may result in "the opportunity to impact the aspirations of civic engagement", “augmenting or promoting civic agency”, and even verging towards critical design, “reading the city as a text or a psychogeographic experience” (Brynskov et al., 2014, 13-15).

In this field, the separation between an urban plan, a service, or a product “is no longer meaningful”, and the focus is on urban social life and communities. In line with visions of inclusive smart cities, UIxD is inherently interested in active citizenship, civic engagement, and democratic and participatory decision-making, putting citizens and communities at the centre of the actor network of the urban ecology (Brynskov et al., 2014).

UIxD embraces sharing economies and the impact of Do-It-Yourself city-making, acknowledging urban planning, design, and governance as an interdisciplinary and collective practice where designed products and services can become platforms for the networked city, leveraging the interdependencies of urban layers and putting under scrutiny the ownership of urban data: for instance, street furniture can be a platform for advertisement, or an online map can become a tool for activism (Brynskov et al., 2014).

Some of the methods to design the city putting at its centre a multiplicity of users are mapping spatial relationships, prototyping in the wild, DIY informal practices and events, and using persuasive design to nudge users towards preferred habits and effectiveness.

CITIZEN-CENTRIC APPROACH

While there is already a record of multidisciplinary references to *citizen-centricity* in the public sector in the design of the smart urban development of digital governance, being the citizens “users, co-creators, customers, consumers or stakeholders” (Sundberg & Holmström, 2024), the concept has no clear definition (Dechamps, 2025; Kamaruddin & Noor, 2013; Lim et al., 2018; Lim, 2023;

Zuiderwijk et al., 2021). The term is generally used along with concepts such as active citizenship and citizen engagement, and similar intentions may be observed in the mentions of a people-centric and a citizen-focused approach (Jakonen, 2023; Cardullo & Kitchin, 2019). It draws from transdisciplinary experiences, frameworks, and methods from (yet not only) design thinking, user-centred, human-centred, universal design, design for all, co-design, participatory design, and agile design, with a focus on collaborating with citizens and communities.

"Tailoring content and services to individual user needs and situation" requires transparency and openness, citizen-driven services (consisting of location-based user-driven personalisation, co-production, and citizen empowerment), participatory democracy, and co-creation of value (Kammaruddin & Noor, 2013). It "allows the citizen to play a role at least at some point in the value chain", whether in the identification of the user, in the design process, in the governance, or in the end result (Dechamps et al. 2025, 3). This approach is seen as a shift from a techno-centric approach to smart cities, to making citizens central in the design of sustainable-oriented innovations (Voelz et al., 2023). In this way, citizens are not seen only as users, but as localised prosumers: as knowledge producers, co-creators, and decision-makers, together with their various communities.

When it comes to communities, they are to be seen "as not something that we develop or design for, but as mutual partners that we design with" (UrbanIxD, 2014, Smyth et al., 2013). While a citizen-centric approach might not require pre-set objectives, it must be responsive to the community (Cardullo & Kitchin, 2019). Nonetheless, an embedded goal of it is to further involve citizens in proposing communitarian projects and creating space for direct participation, also when approaching complex and wicked problems.

Practitioners concerned with urban futures should be ready to address the multiplicity and intersectionality of citizens' identities, expressions, and needs, acknowledging their role in how the city and its users cohabit and influence each other in citizenship. Hence, whether it is designing for inclusivity and accessibility for wider audiences or providing more tailored or customisable solutions that can feel personal and relevant to specific target groups, social cohesion is in the hands of urban designers and placemakers. This also means designing for what is often defined as "third places" (Oldenburg, 1989), spaces that create social environments outside of the users' home or workplace.

Here, technologies (together with people and places) become opportunities to explore relational approaches for an active role of citizens, with methods such as citizen sensing (Gabrys, 2017; Ponce et al., 2025) and citizen development (Voelz et al., 2023). Citizen-centrism may seem strongly intertwined with smart citizenship and technologies, but it may also contribute independently, as a value and methodological approach for UIxD. It invites an intersectionally inclusive and participatory approach to design for, with and by (Sanders, 2002) citizens, and their interactions with the city and its technologies, giving citizens some technological sovereignty and making digital infrastructures a public-common (Cardullo & Kitchin, 2019).

If the understanding of citizenship is then critically extended outside of its anthropocentrism, the use of technologies in the urban space should also consider the rights, needs, wisdom, temporalities, and interdependencies of more-than-humans to avoid or limit human contributions to (if not mitigate or even fight) "challenges associated with high-density living, poor resource management, habitat loss, and pollution" (Clarke et al., 2019). Projects interested in environmental sustainability can simply not be human-centred and technocratic. While this project focuses on social sustainability, it may benefit from a perspective on animal-computer interaction. This approach would help locate an artefact, service, or experience in a bio-inclusive actors map and foresee how it can impact the lifecycle and maintenance of the design, beyond its impact on wildlife.

Clarke et al. (2019) also suggest drawing best practices from non-hegemonic wisdom and decolonial practices while finding inspiration in Indigenous epistemologies and ontologies to develop a different understanding of “smartness” for collaborative urban futures.

URBAN INFORMATICS AND UBICOMP

In a time of hyperconnectivity, while our general dependence on ICTs can be the object of criticism (Brubaker, 2022), their role has become such a defining cultural phenomenon that it has shaped a new era of media, described as Gutenberg Parenthesis (Pettitt, 2007). The potential of the ubiquitous computing technologies is now studied by academia to be understood and designed also in its own field – ubicomp (Weiser, 1994). It is “to be picked up, adopted, adapted, and appropriated to personal and collective goals” (Dourish & Bell, 2011), with a critical lens on ever-present technological artefacts “as embodiments of sociocultural theory" to discuss their affordances and their possible impact on society. Urban informatics explores the opportunities unlocked by ubicomp and pervasive design within the city, studying, designing, and mediating digiphysical urban infrastructures between people, places, and technologies (Foth, 2008). Focusing on participatory urbanism and citizen-centric applications of urban technologies and media (Barns, 2014), it aims at hosting space for citizen-ability (Foth, 2018). This is where Foth locates smart citizenship: in the interdisciplinary meeting between civic activism, social media, and media architecture. Similarly, platform urbanism also works within the impact of digital platforms on urban services and space (Barns, 2014).

The widespread adoption and application of products of such advanced engineering made them everyday accessories. However, they are more than gadgets of smart lifestyles: they influence the “legibility” of the city – the epistemologies, ontologies, behaviours, and meanings citizens develop around urban spaces. This notion acknowledges that technologies and networks give shape to elements of our social lives through data, systems, and interfaces. Consequently, devices and platforms become the definitive means to negotiate them. “Digital technologies that render the world legible in new ways” work in situated contexts, and “local legibility” reflects how people engage with their local reality, also “grounded in Indigenous practice” (Dourish & Bell, 2011).

Similarly, “literacy” and “legitimacy” are other important elements associated with ubicomp, useful to reimagine users’ relationships with tech. The first is connected to “informativeness” and “representational practices”, admitting the role of ubiquitous technologies in knowledge production and the importance of them embedding and being embedded in different cultural experiences. The latter identifies the connection between digital systems and users’ understanding of an environment as result of actions and habits – creating the risk, however, to fall into audit culture and virtualism (Miller, 2003 & Strathern, 2000, as cited by Dourish & Bell, 2011), aligning reality with formalised models, moving “from being ways of describing the world to being ways of organizing it”.

“Inextricably intertwined with social exclusion" (Brynskov et al., 2014), the digital divide (van Dijk, 2006) is then a result of design. From the digital experience may derive loneliness, polarisations, and radicalisation, so disconnected subaltern urban and digital forms of citizenship are to be considered, as well as digital barriers (Soares de Oliveira Neto & Kofuji, 2016) and media representation. Urban digiphysical infrastructures may have a role in digital democracy by designing the city as an interface against the privatisation of technologies and spaces and technofeudalism (Varoufakis, 2023).

AUGMENTED URBAN MEDIA ARCHITECTURE

From the first neon signs, through the *Spectacolor Board* in Times Square in the 70's and 1994 Amsterdam's De Digitale Stad – until today – media architecture and media façades has been evolving to become always more situated, resulting also in Architectural User Interfaces that engage with people and community (Foth, 2018), while creating possible critical spaces (Colangelo, 2019). With technological advancement, reactive and interactive media and the urban digital layers eventually paved the way for immersive devices and experiences for hybrid spaces within the city.



Fig.1: street lighting creating night parking lots for bikes (Amsterdam, 2025)

From Wheatstone's *stereoscope* and Heilig's *sensorama* for a multisensorial "experienced theatre" to today's headsets and mobile augmentations, there has been development in the exploration of immersive experiences of enriched and parallel perceptions, visions, and worlds in what is Extended Reality (XR). This works within the Virtual Continuum and encompasses a wide spectrum of experiences (Milgram & Kishino, 1994), ranging from video displays, light design, and near-reality augmentations to fully virtual and immersive environments supported by an alternative world of computer graphics.

Other than its most common applications in gaming and entertainment, it finds different industrial and commercial uses, as well as applications in research and art (Bimber & Raskar, 2005), but XR and Mixed Reality (MR) also offer opportunities for applications in cities, including in city planning, urban regeneration, data visualisation, education, information and sensitisation campaigns, e-governance, and more. For instance, Urban Digital Twins and digital city models found uses within strategic urban development in Augmented Urban Planning (Azadi et al., 2025; Sassmannshausen et al., 2021). Nonetheless, interactions in spatial immersion, experience simulation, and virtual spaces are also useful means for social ecological urbanism (Augmented Urbans, 2025).

In the wide choice of opportunities and technologies, towards the virtual edge of the spectrum, more resources and technical requirements are needed to partake in an experience. A fully Virtual Reality (VR) becomes less accessible, requiring specific devices and digital 3D modelled realities. Consoles, immersive exhibitions, and, only rarely, MR arcades dictate a mostly private and individualistic use of these technologies. In the meantime, the opposite side of the spectrum of the Virtual Continuum – near-real Augmented Reality (AR) may set a common access to the digital layers of the city while questioning reality itself.

Some of the main commercial examples of these technologies involve the use of mobile phones: reality is captured through the camera, processed by software, and finally augmented on the screen. Niantic's Pokémon Go is a popular app that extended access to XR on a mass scale. However, while some of its features address AR, as through the camera lens, most of its system is based on Augmented Virtuality (AV): creating a virtual twin of the world to be explored through the physical body that is reduced to a piece of the interface. Widely accessible forms of AR can also be found in social media, through the use of camera filters: by overlapping digital layers to captured images, users and influencers can edit their photos through filters often created by other users. Another popular exploration of the Virtual Continuum that verges towards real environments can be found in city "portals": real-time video screenings that aim at connecting different cities worldwide, yet not without ethical and technical challenges (Forbes, 2024). Similarly, interactive urban signage, which may incorporate live data from sensors around the city, offers another example of near-real augmentation (yet far from virtuality).

XR, MR, and AR also develop through digital projections, in public and commercial services, as well as in entertainment and artivism. Brands and stores use gobo projectors to advertise their logo and services; projection mapping is increasingly common in public events, animating buildings of the city centre; and guerrilla projection is providing new transportable and ephemeral ways of microvandalism to share sociopolitical commentary. Light art, video mapping, facade projection, and other types of media architecture can offer access to XR, but this thesis will specifically focus on how visual projections and projection mapping may contribute to inclusive smart cities through a citizen-centric approach.

SPATIAL AUGMENTED REALITY AND AUGMENTED PLAY SPACES

Places "where playful behaviour and interaction emerge, facilitated by a technologically and sensory enriched, generally accessible environment" define Augmented Play Spaces (APS) and allow users into an experience through inviting interfaces in a public context (Mast et al., 2021). Where the space itself becomes the arena for virtuality, Spatial Augmented Reality (SAR) conceals opportunities to free users and citizens from their devices, for their interactions to be redirected to the city (Bimber & Raskar, 2005).

As Cooperative Tetris (Mast & De Vries, 2016, as cited by Mast et al., 2021) [Fig.2] can exemplify, APS conceal opportunities for social interactions, whether through the use of screens, as in this case, or through more advanced solutions. Regardless of the technological approach, UxID and Media Architecture have the power to engage citizens, as Theatre For You (Barnard et al., 2021: 71) [Fig.3], that uses led lights for playful placemaking.

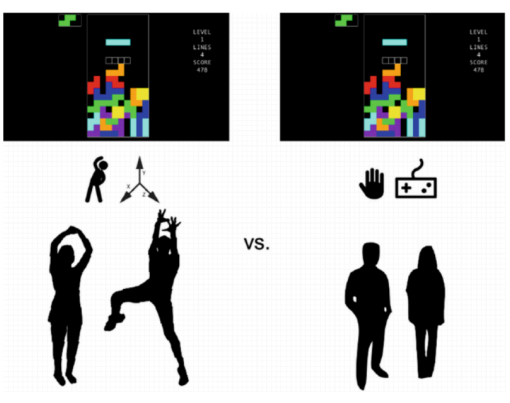


Fig.2: Cooperative Tetris

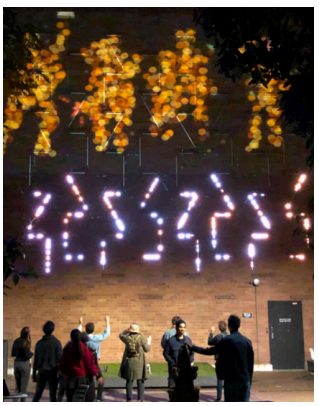


Fig.3: Theatre For You

MALMÖ

By 2034, the Municipality of Malmö is expected to reach 400 000 inhabitants, and the city is progressively changing in that direction (Malmö Stad, 2024). Some of the core areas for this growth are already evident – such as Hyllie, Limhamn, and Västra Hamnen, but this process will also involve less frequented areas and more peripheral places, like Jägerso and Rosengård.

This population growth is a key element in the development of the city and may confirm cultural diversity as part of today's identity of Malmö and its citizens, as it counts on circa 188 nationalities. Designing for the needs of immigrants is increasingly important, and it can benefit from a deeper understanding of urbanisms in the so-called "Global South" or "Global Majority".

The role of design and technologies in social interactions (as in social media, networking, gaming, etc.) online and in the city becomes evident as they bring people together (or do not). The joint use or sharing of a space, an artefact, a technology, a service, a network, or a medium may ceil potential for community-making, to be explored in this project with a focus on citizens.



Fig.4: Ljudkullen in Scaniabadet



Fig.5: Area in Pildammsparken hosting speakers and a light show

In the context of the city of Malmö, there are already local case studies that use technologies and media to create space for citizens' expression: for instance, Ljudkullen [Fig.4] is a place where the people can take control over an experience by playing their own media, with the permission of the municipality (usually requiring a payment). A similar technology is also hosted in Pildammsparken, where a light show gathers nocturnal crowds [Fig.5]. An analogue experience worth mentioning is the Vänskap Bänkarna - benches against solitude [Fig.6], as well as the interactive kid's game in Folketspark [Fig.7]. Then, urban media less focused on community are "Elsewhere" by T. Ruiz Gutiérrez (a permanent installation in Malmö C) [Fig.8], "Spårsken" by C. Partosthe (a sound-reactive light installation in Triangeln), the new projections in Friisgatan [Fig. 31], and advertising screens around the city.



Fig.6: Vänskap Bänkarna



Fig.7: Interactive kid's game



Fig.8: Projections in Malmö C

METHODS

Drawing inspiration from Design thinking (Brown, 2008), Human-Centred Design (IDEO, 2015), and Universal Design (Connell et al., 1997), a participatory approach to User-Centred Design (Norman & Draper, 1988; Sanders, 2002) within the city is here object of study. Putting the citizen at the centre of a reflective and iterative practice (Schön, 1987), different methods are used to feed this process of design research (Cross, 1982; Frayling, 1993; Stappers, 2007; Zimmerman et al., 2007).

ENGAGING WITH EXPERTS

The project involves experts from academia to inquire into different geographies and urbanisms, for a culturally aware approach. In this case, the Network for Global Southern Urbanisms supported understanding the concepts of urban and citizenship in a cross-cultural way, with awareness of different sociocultural, ethnic, and geographic experiences, urban traditions, and situated needs. During the seminars of NEGSU, 2025, and thanks to their collaboration during their interviews, the designer engaged with some of the work and ideas of Wangui Kimari, Madina Tlostanova, J. Miguel Kanai, Tariq Jazeel, and Kavita Ramakrishnan.

Later in the process, an interview with landscape architects at Malmö stad also contributed to understanding the challenges of designing with light and testing in public spaces, leading to some of the conclusive considerations on the feasibility of the final design.

DESKTOP RESEARCH

A designerly and systematic approach to desktop research and online sources is used here to browse through existing literature and case studies, building a theoretical overview presented above, while surveying good practices and designed experiences, services, and artefacts, together with the technological affordances and requirements.

CONTEXTUAL OBSERVATIONS

Observing urban interactions of people in and with public spaces in three different squares built a situated inquiry on their stationary and transitory presence, and on seating and standing. The context of the square is constantly negotiated through citizens' interactions, and studying them is a way to account for them when designing public experiences (Dourish, 2001).

DEMOGRAPHIC DATA VISUALISATION FOR MAPPING AND ZONING

The demographic data is used in this project to map and visualise the hotspots of population growth in the city of Malmö between 2024 and 2034. Dividing them with a colour system, the most inhabited areas, the ones with the highest population growth, and those with significant population growth after 2029 are then represented. Adjacent areas become visible, and the designer identifies possible connections and collaborations by zoning the urban. In this case, this is used to group some areas by vicinity to pinpoint possible spaces to host the design research and final proposal.

(CITIZEN) INTERVIEWS

Approaching the set target to inquire about their urban experience, citizens soon become the protagonists of a user-centred and human-centred project, where their needs, abilities, and behaviour are objects of inquiry (Norman, 2013). Semi-structured interviews with citizens from

different cultures are conducted to understand their perception of public spaces in Malmö (as squares and parks), the behaviours and activities they would notice or enact (also in comparison to their previous experiences of similar spaces in their native countries), the challenges and opportunities, and their needs, hopes, and wishes for the urban futures. Eight participants were involved: four face-to-face with a coffee or some water, and four via phone call. Interviews lasted between 30 and 40 minutes, and were led in their preferred language between English, Spanish, and Italian.

By answering questions about their relationships with the city, they supported shaping cross-cultural observations on the urban spaces of Malmö. For this project, the citizen data to be inquired are age, gender, nationality and cultural identity, the district they live in, the reasons that brought them to Malmö, their occupation and hobbies, the places they visit and the ones they do not, their challenges and struggles, and the general perception of the public squares and parks of Malmö, and finally, more explicitly, their needs, wishes and hopes.

(CITIZEN) PERSONAS

Personas are used to establish a better understanding of the target (Pruitt & Grudin, 2003), and to then involve them with a participatory approach. The inclusion of different diversity dimensions (Himmelsbach & al., 2019) helps inspire a baseline definition of citizen personas as an intersectional vision of citizens. Unless stated otherwise, a citizen is here considered a person living semi-permanently in the city but potentially from any place in the globe, of any ethnicity, culture, religion, any appearance, gender, age, or sex, with or without any ability, resources, knowledge, occupation, interests, or community.

A situated target is here explored, with their experience of citizenship and their relationship with the urban. Relevant factors set for this project are age (20-34), a shared language (English, in this first iteration), employment and hobbies, and urban areas of interest.

CO-IDEATION WORKSHOP

The project sets a collaborative workshop to share the decision-making process with a small poll of volunteer citizens. A participatory approach may pave the way to a democratic process and result, and to mutual learning and the shared production of knowledge (Bødker et al., 2021).

This consisted of three parts: a focus group on different sharing spaces for intercultural dialogue, an individual sketching session to envision an outdoors sharing/meeting place in the form of a public square, and a joint discussion over potential XR experiences for community-making.

REFLECTIVE REFRAMING AND ITERATIONS

Reflection-In-Action and On-Action (Schon, 1987) supported an iterative process of knowledge production, evaluating implications, consequences, and new opportunities during the activities, allowing adaptability and responsiveness to citizens' needs, fears, and contributions.

SKETCHING AND PROTOTYPING

Sketching – shaping thoughts into ideas through words and drawing (Buxton, 2017) – is used in this project as a way to equip the citizens as co-designers. Starting from the results from this practice, the designer develops them further and refines the concept through prototypes, defining and testing the research artefact to reach their evaluation (Sanders & Stappers, 2014).

PARTICIPANT JOURNEY MAP

Mast et al. (2021) propose a model to design a motivational non-linear user journey within an Augmented Play Space, and it is used here to describe the experience tested for the concept. This includes six potential states: transit, awareness, interest, intention, participation, and finishing.

TESTING

Challenging the design concept in real scenarios and with the users – in the city, and with the citizens, in this case – is necessary to understand the weaknesses and failures behind it (Norman, 2013). Testing different parts of the design artefact and doing it repeatedly supports its improvement, while leading observations and conclusions to emerge.

PROCESS

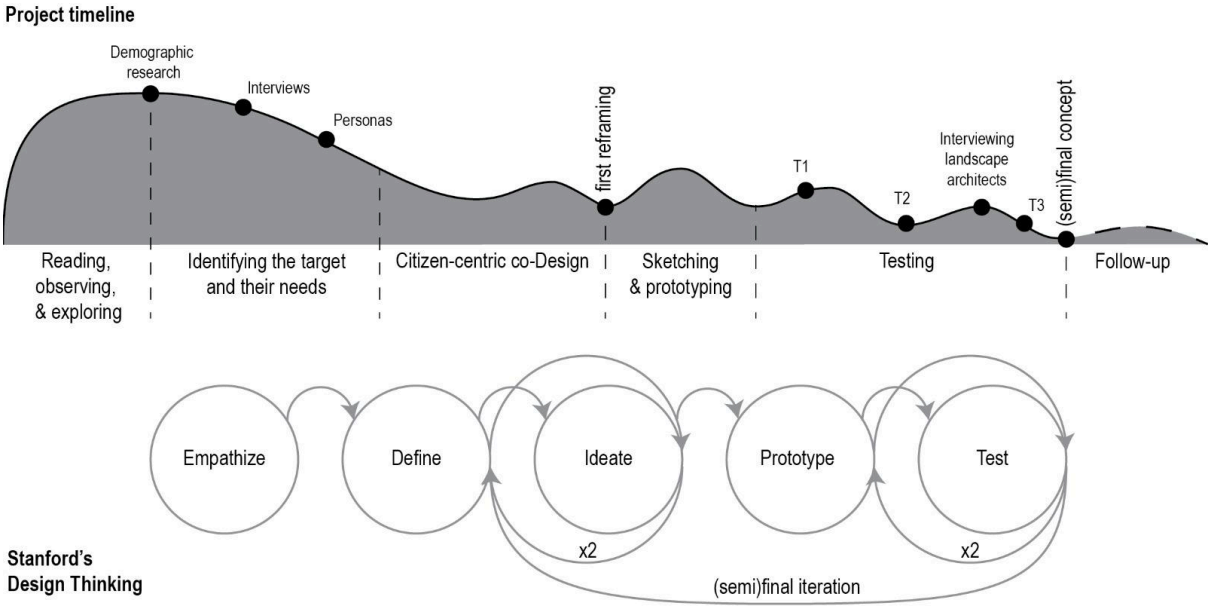


Fig.9: project timeline representing divergence/convergence through time and iterations within the Design Thinking framework

The project was initially based on a rooted understanding of design frameworks such as Stanford's Design Thinking, IDEO's human-centred design and the UK Council's Double Diamond. Key elements such as empathy/discovery/inspiration, definition, ideation, sketching, prototyping, and testing led the process, but, as citizens got more involved in its different phases, this adapted to prioritise their contributions and insights, resulting in repeated iterations and reframing. This resulted in a concept to be considered final within the scope of this project, but to be further explored and adapted for its eventual real development and implementation within situated urban communities.

RESEARCH

The project started with the acknowledgment that urbanisation and digitalisation are not designed for everyone's cultural expression, representation, or sensibility (Michieletti, 2024). This inspired the designer to explore how cities may design augmented/ing media (such as projections) to make smart cities more inclusive and support urban community-making. Through the research phase, with the identification and involvement of the target and after the data gathered on their needs, hopes, and expectations, a clearer research question emerged. While the project reflects on the role of the citizens in the design process, the democratic access to XR, and some of the challenges of designing with light, it seeks to answer:

How may projections augment public spaces to facilitate cross-cultural dialogue and mediate social inclusion and cultural belonging?

SEMINARS WITH EXPERTS

To develop a cross-culturally inclusive understanding of the urban, the research phase started with an exposure to experts on cities and urban design from other geographies and cultures, attending the seminars of the Nordic Network of Global South Urbanisms at Malmö University. Supporting their team as a videographer during the interviews with the speakers created chances to further engage with them. This resulted in a wider critical perspective on the urban spaces of Malmö.

The acknowledgement of different elements from some Southern and Eastern placemaking practices would feed into the discourse later developed with the multicultural target of the project, with the goal of building a culturally aware approach to smart cities of today and tomorrow.

"*The urban ultimately is people, and the people who hold up the urban are often the ones excluded from the services that it offers*."

Wangui Kimari (NEGSU, 2025)

PRELIMINARY FIELDWORK

Initial contextual observations were conducted in three different spots in the city centre of Malmö (Stortorget, Adolf Gustav Storget, and Smörkajen) to shape tentative assumptions on how people interact with and within public spaces such as squares.



Figure 10



Figure 11



Figure 12



Figure 13

Citizen's approach to squares here considered are: stationary (staying for more than a few minutes – e.g. resting or waiting for somebody) and transitional/consequential (when the square is

just a place to pass through, and when citizens' presence is a consequence of their engagement with other stakeholders in the surroundings – e.g. shopping or eating there). People's intentions may, however, quickly change as their use scenarios and goals are only temporary.

A break with sedentary rest appears to be one of the many motivators holding users in squares [Fig. 10 & 11], as deductible from the fact that most benches tend to get occupied (but not often shared with strangers). Depending on the weather, another intention may be sunbathing looking for shelter (whether there are walls or roofs that can protect from rain and wind). Meeting someone is another reason that brings people to the square. As a landmark, it also seems relevant in the wayfinding of the city.

Some other common reasons that are assumed to bring people to squares – even if not directly noted from these observations – can be public events, demonstrations and speeches, and entertainment and performances. Architecture and monuments seem to capture much (media) attention, with people stopping by or purposefully going to take pictures or videos – presumably tourists [Fig. 11].

People on their own often have their smartphones in their hands and wear headphones [Fig. 10]. Individual users and small groups stationed in the squares show a preference for sitting. Youths and bigger groups would mostly stand or lean on surfaces. Slow walking is also a common choice, which keeps people in the square for a longer time.

A second round of observations was carried out later, during Malmö Festivalen, once the target and design opportunities were set, to record data from a different setting: a crowded night, potentially with fewer foreign tourists, but more regional visitors. Some of these are presented in the results of the last testing session.

DEMOGRAPHIC ANALYSIS

The demographic analysis helped understand which citizens and districts to focus on. Reports from Malmö Stad (2024) indicate that around a third of the population in the Municipality of Malmö was born abroad, with 187 different nationalities present in the city. Among the most numerous are people from Iraq, Syria, and Denmark. Most immigrant people getting to Malmö are aged 20-25, while most people moving out are 30-45 (e.g. families with children, often moving to nearby municipalities). People under 35 make up to 47% of the total population, with 18% aged 25-34.

Today, Malmö is the fastest-growing city in Sweden. With an increase of population of 1% in 2024, this is expected to grow by almost 10% before 2034, reaching around 400 000 inhabitants (Malmö Stad, 2024). Some of the areas currently developing to become the core of the population growth are: Limhamns Hamnorade (+6%), Västra Hamnen (+20%), and especially Hyllievang (+85%). Other areas with considerable prognosticated growth are Inre Hamnen (+600%), Kalkbottet (142%), Norra Sorgenfri (+134%), Vintrie (+71%), Flensburg (+67%), Ellstorp (+34%), Elinelund (+32%), Kirsenberstaden (+24%), Holma (+18%), Bunkleflo (+15), Tornrosen (+15%), Värnhem (+7%), and Segevang (+7%). Between 2029 and 2034, some other areas are going to see an exponential increase in their population, such as Kvarnby, Fortuna Hemgården, Jägersro, Rosengård, Frihamn, Mellersta.

MAPPING AND ZONING

Based on the data on the population growth per district by 2034, the areas with the highest expected growth have been identified to help define some hotspots in the city [Fig. 14]. These areas should raise attention, considering that some count on the highest number of inhabitants (red), and others with the biggest growth percentage (orange and yellow). Interpreting these data led to a visualisation of these geographical areas, which helps zone into adjacent areas that may benefit from being reconnected to the urban fabric.

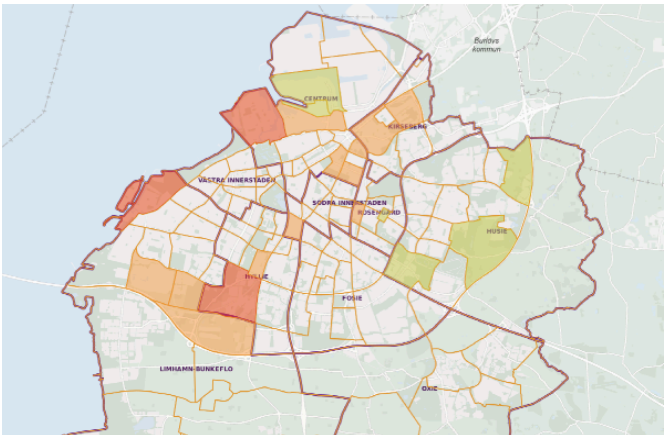


Fig.14: Demographic growth in 2024-34

INTERVIEWS

Considering the role of immigrants in the city, semi-structured interviews reached out to local citizens who have been in Malmö for less than three years, originally from Syria, Zimbabwe, Angola, Bulgaria, Germany, the UK, Chile, and Brazil. Most moved to study, two for a PhD, and one as a freelancer with their partner. Five identify as women, two as men, and one as a non-binary person. Only one of them speaks Swedish.

They all agreed to share their experience in urban Malmö as immigrants living public spaces in relation to their own cultures and countries. Their interviews shared valuable insights that helped shape some of the needs to be further addressed by the project:

- Friends and acquaintances, cultural events, or businesses and shops are the main reasons for bringing people to different areas of the city. In the absence of motivation to visit them, some places remain unknown to many, even after years of living there. This is seen in different areas in urban and peri-urban Malmö.
- Without organised support such as that of the university ecosystem or a workplace, socialising can be challenging.
- In Malmö, public spaces such as squares are described as quiet and individualistic; in other cultures and geographies, people may engage in social interactions more easily (even when superficially) and spontaneously enact group activities open to all. There is a shared feeling of judgment and cultural inappropriateness that stops people from feeling free to run, jump, or dance, being loud, and expressing more openly and soundly in local public spaces.
- Elsewhere, community-making is facilitated by cultural centres and neighbourhoods, promoting opportunities for entertainment (e.g. concerts and movie screenings), sport (e.g.

dance, soccer, basketball), and more, unorganised and free of payment or registration. The price of joining any activity is an economic barrier repeatedly mentioned by the interviewees.
- Food also plays an important role in urban social lives and communities. Cooking in public spaces, eating together, and sharing food with others bring people together. Food courts also activate and populate public places and get crowds to interact.
- Green areas represent great value for the city, but they are seen as a utility, rather than an opportunity to connect with nature
- A general feeling or preconception of danger may represent another obstacle for people to engage with different urban areas of Malmö (especially for women alone by night).
- Some of the needs and hopes expressed pointed to more cultural and sports activities free of charge (especially in winter and at night), and better lighting in different areas of the city.

CITY PERSON(A)S

After setting immigrant citizens aged 20-34 living in Malmö as the main target of the project, a brainstorming session on the many possible personas that make them up led to a few representative main ones:

- A person in their mid-20s who moved to study in Malmö. Their friends are mostly international students, and they go to visit them in different areas of the city. They hang out in the centre but prefer parks and green areas. In the evening, they often go to Möllan or join a house dinner/party. They do not speak much Swedish.
- A person in their late 20s/early 30s who is seeking a job to stay indefinitely in Malmö. After some months or years here, they already know the main spots in the city, so they are always looking for something new or unknown to them. They are interested in building a network and a community. In the evening, they might be coming back from an event or from a friendly meetup. They must be mindful of their expenditure until they get a job. They want to learn Swedish, but they need stability first.
- A person who moved to Malmö through (or to) work, in their early 30s, with a social circle rotating around their job and possibly hobbies or sports. They are building long-lasting relationships. In the evening, they might be going to a pub in the centre or going back home from a late shift. They are currently trying to learn Swedish.
- A person who moved with their family seeking asylum. They may have initially lived elsewhere in Sweden but moved to Malmö to study and look for career opportunities. They like people and parties. They are already fluent in English and Swedish.
- A person who moved to rejoin their family. They live in the periphery and are currently unemployed, so they do not go to the city often. They do not speak English or Swedish.

It is important to note that any of these may belong to a non-hegemonic group, in connection to their culture, ethnicity, gender, ability, religion, sexual orientation, socioeconomic class, or political ideology. What most of them have in common, other than their city and age range, is a general struggle learning the local language and integrating into Swedish communities.

IDEATION

CITIZEN WORKSHOP

A co-design workshop to share the decision-making process with a small poll of volunteer citizens was conducted in the facilities of Malmö University. To offer value for their time, the workshop was framed also as an intercultural *fika*, and coffee, pastries, fruit, and other snacks were provided by the organiser. The four participants were from Germany, Poland, Bulgaria, and Angola/Italy – three of them attended following up on the interview.

The activity was divided into three parts. The first created different settings for intercultural dialogue, to explore ideal spaces for sharing through colloquial conversations, concluding in a group de-briefing session. The second consisted of an individual sketching session of 20 minutes to envision an outdoor sharing/meeting place in the form of a public square. As an optional challenge, they were invited to consider night, cold, wind, and rain as possible constraints. The final part was a collaborative session to make ideas come together and design a possible approach to the projections in the city.






Fig. 15, 16, 17, & 18: the four different sharing spaces explored in the focus group

After a welcoming introduction to the project and workshop, the participants were led into three five-minute sessions to get to know each other, chat, and share in couples, different at each turn. Each time the participants were invited to pair up, they chose together one of four sharing spaces: one where they could freely sit on the ground (with a carpet and some pillows) [Fig. 15], one with two chairs facing each other [Fig. 16], one with two chairs in perpendicular positions with a table between them [Fig. 17], and a small sofa [Fig. 18]. The goal was to observe whether different settings can influence a conversation and how they can create comfort in quick social interactions. To facilitate this, they were invited to consider thinking in advance about something to share: a story, a cultural activity or artefact, an artwork, a song, a dance, a game, food, or anything they would like to propose – whether material or intangible, from their culture or simply from their own liking. Three decided to share stories, while one brought chocolate from her country. These exchanges would become icebreakers during the first part of the workshop.

Nine different conversations resulted in individual private moments of reflection, which were then shared among the participating citizens. The topics discussed included: their experience in Malmö, urban centres, public spaces, cultural events, cultural exchange, and sharing memories through multimedia, recent news, and gossip.

Two people preferred sitting on the ground, but one did not even consider it: clothing and limitations of physical mobility can become a barrier to doing so. The other two preferred sitting face to face as it allowed “to create eye to eye contact and a more direct connection”. The table did not come as support to these interactions.

Fika was cherished, but a participant mentioned that food is a more suitable social stimulus for bigger groups, as eating or chewing can be perceived as rude in a conversation. The initial indication

to prepare something to share was deemed useful support to kickstart and direct the activity. Visual aids, such as digital photographs and videos, were used by someone to support their storytelling.

During the de-briefing sharing circle, the citizens expressed a common feeling of being foreign to a new city, an interest in creating community and networking, and a need for free public spaces and cultural centres to socialise and meet other people. Talking and “yapping” seem the main ways to do so: a participant citizen notes that inviting someone to eat or drink something is usually just an excuse to meet and chat. They all expressed interest in continuing the conversations.

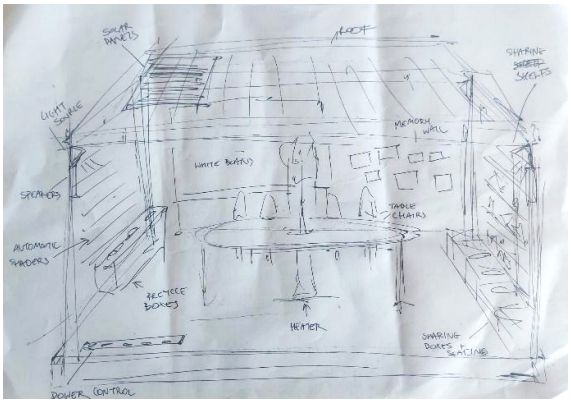
When questioned about the role of technologies for community-making and meeting others, they agree on it being saturating and superfluous (or even negative, when discussing certain social media platforms). Nonetheless, they acknowledge it as useful to reach out and communicate meetups or to share multimedia. However, they refer to the reactions to the 2025 Iberic *apagón* to remind about the importance of creating tech-free space and time to reconsider traditional cultural practices (e.g. arts and craft, music, dance, and cuisine) for community-making: among the consequences for the digital amputation resulting from a blackout lasting over twelve hours, Spanish streets and squares got crowded with people socialising offline.



Fig. 19: Projections during the de-briefing

Different visual projections on the ground were screened during the second part of the conversation [Fig. 19] to introduce the technological approach studied in this thesis. Images in motion from an online video library supported that: a blooming field, a sandy desert, a biodiverse ocean, and a scene of fluid abstract colours were shown, at their feet. As they were asked about how these influenced the de-briefing session, they expressed appreciation for grass and flowers: one of them “wanted to sit on the ground”. The colourful patterns were distracting, instead.

During the second part, the participant citizens drafted squares [Fig. 20 & 21] that could provide the facilities to host public outdoor third places, considering constraints such as night, wind, and winter. Another group moment would later allow everybody to share their thoughts and proposals. Some elements that were mentioned are: greenery, ceiling/roofing, solar-powered lights and outlets, heating, seating, a table for collaboration, sharing shelves, creative supplies, sport equipment, games, appliances to play music, space to dance, modular and multipurpose urban furniture, a customisable digital wall, a restroom, and somebody that takes care of the whole facility.



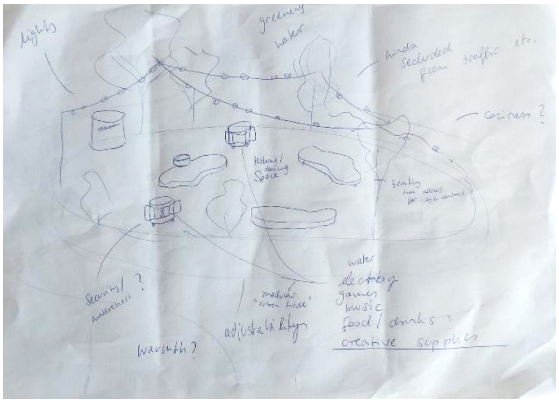


Fig. 20 & 21: Sketches from two citizens proposing different settings to facilitate sharing culture in public spaces

The third and final part of the workshops tried to embody with paper a small square in the room (where projections had been previously showing), for the co-designers to approach it as a white canvas and furnish it with each other's ideas and thoughts. In addition, they were introduced to some of the technical opportunities behind audiovisual projections and SAR, inviting them to include them in their solutions, in what later would get named "Video Stations". Four possible screening settings were explained: projections on the ground (or ceiling, or a wall), on one wall and the adjacent ground, on a corner, and on urban furniture or holographic displays [Fig.22]. They decided to explore the first setting: projecting on the floor. This immediately took the shape of a streetlight, as it is already a real and common piece of urban furniture that could be repurposed.

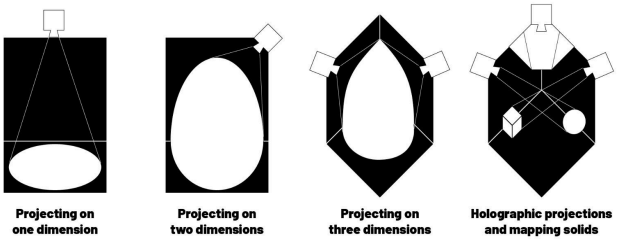


Fig. 22: Some projecting settings

Whereas a paper canvas was ready to host the sketching, the whole session developed into a co-ideation conversation while sitting in a circle, rather than collaboratively crafting on the floor. Some of the proposals for "Augmented Squares" further discussed were projected sport courts, digital toolboxes, and a memory wall to leave citizens' traces. Then, one of the participants eventually pointed out that what they would most appreciate are icebreaking activities that could recreate the relational feeling from the first part of the workshop, finding consensus in the other attendees of the activity: in this proposal, an opportunity is identified for the city to *"design people out of their screen's individuality, by making the square a shared screen itself"* for citizens to meet on, as rephrased by one of the participants. Hence, the citizens finally agreed on the proposal of placing conversation starters in public spaces to get citizens to come together through projection mapping. A "*yapping station*" to meet others and to share culture, knowledge, memories, and opinions.

RESEARCH REFRAMING

By involving the target through interviews and a co-design workshop, the process aimed to include the citizens in ideation and decision-making, facilitating activities to collaborate towards the (semi)final design concept. A reflective approach brings this forward and iterates on it, evaluating challenges and potential.

How may augmented squares bring us together?
How may projections support interpersonal dialogical interactions?

The workshop led to the idea of Augmented Social Streetlamps, projecting different icebreakers around the city to make more places to meet others and build community, while activating different urban areas through a platform of augmented services and experiences. How to reimagine public streetlamps as interactive urban furniture to create ephemeral third places seems worth exploring as a way to mediate cross-cultural dialogue, inclusion and cultural belonging. Among the icebreaking activities considered for this interactive service, those requiring previous knowledge

or capabilities (such as in singing and dancing) or additional resources (as in food for *fika*) were ruled out. Allowing the project to deepen only one proposal in the variety of possible experiences, the concept is set to projecting visographic icebreakers to kickstart a conversational experience, promoting dialogue, cultural exchange and active citizenship, and potentially tackle loneliness and polarisation. The initial media proposed by the citizens are questions to break the ice, such as “What’s your best memory in Malmö?” or “What’s your favourite place in the city?”. In this way, the city of Malmö is not only the context, but both the object of and the platform for exchange.

SKETCHING & PROTOTYPING



Fig. 23: A digital sketch (and digital collage) of possible experiences such as playing, chatting, and dancing.

Starting from the citizen’s idea, the sketching phase opened the process to divergence again, initially envisioning “icebreakers” also as non-verbal experiences, revealing vast space for development [Fig. 23]. The wide range of opportunities may be motivational to further explore the open concepts of Augmented Squares or Video Stations, hosting different interactive and immersive experiences, such as interactive bulletin boards, videobenchs, tele-speaker corners, etc. However, the semi-final concept focused on using literal questions as conversation starters, as suggested by the citizens and co-designers.

TESTING

Initial lo-fi prototyping explored both the interface and the experience of the service, focusing on a three-steps process: participation (activation), meeting, and conversation. Three different testing sessions helped analyse some of its basic elements and of people's interactions with and within visual projections in public spaces. The sessions led to different iterations and adaptations that would inform the final concept, feeding on a research-through-design approach useful to test initial assumptions and validate the idea.

TESTING THE INTERFACE AND CTA

During the first test, a button with the call to action "Wanna chat?" was set in the hall of the university (a semi-public space, with similar elements to squares). A roll-up inviting others to "JOIN THE CHAT" was set to open when somebody pressed it, but, while the button caught much attention, no one pressed it. Different elements may have contributed to the result, but this can be interpreted as a challenge in getting users to interact with an experience through manual activation.



Fig. 24: Testing social interactions after somebody has already initiated an experience

Consequently, the Call to Action was also tested within an already active experience [Fig. 24]: the button and the sign were placed in the hall of another university building (another indoors public space), and this time the call-out roll-up was already open with a person next to the button. The conversation would then be nudged through an ice-breaking question: "What is your favorite memory in Malmo?". This resulted in four different conversations within 30 minutes (which involved only non-Swedish users), suggesting that an active (social) experience is more engaging than one which requires its manual activation.

A projector was initially set to complement this activity, but the challenges engaging users (and the lack of a formal permission to proceed with projections) led to focus on the components separately, to first understand what role would a button and a sign have in the interaction (as in turning on the streetlamp or playing the media): the result was that the experience should be running freely and the media should be already playing, without the need for an extra step for activation.

TESTING CITIZEN'S INTERACTIONS WITH PROJECTIONS

The second phase of testing addressed possible citizens' interactions with a visual projection outdoors and by night. To design around limitations such as the lack of access to electricity or formal permission for the use of public spaces, the testing involved the use of a lo-fi prototype made up by a flashlight and a plexiglass stencil with cut-out letters stuck to it for shadow play [Fig. 25, 26 & 27]. This Wizard of Oz test had the goal of observing how a visual projection may influence how citizens perceive and approach public space, and what types of engagement it would get.



Fig. 25, 26 & 27: Wizard of Oz test of an example of projected icebreaker for the Social Augmented Streetlamps

Observations resulting from this testing phase are:

- Projections create a perception of defined space in many: most people would move around it or stop in front of it, not daring to be the first to step on it.
- Pictograms were deemed useful by some users, so iconography may support a more universal approach by creating a more immediate semiotic interpretation.
- The format of the Call to Action can be only partially evaluated, as there was no way of tracking its long-term impact. Even if it was read in the immediate moment, there is no way to know how it influenced the users' conversations during that night.
- It might be assumed that the experience required additional features to contextualise it and explain how to engage with it for the first time.

During the test, the main interactions observed were:

- visual engagement (staring at it and reading through the text).
- bodily juxtaposition (walking through, jumping on it, and moving around).
- gestural engagement/mixing (trying to "touch" the elements of the projections and moving in relation to the media).

No conversations were started directly within the perimeter of the projection, but this may be attributed to the constant and immediate presence of the designer and to the context of the testing: a crowded festival full of free cultural activities (whereas the project was proposed to complement their absence).

TESTING CITIZEN'S INTERACTIONS INSIDE PROJECTIONS

As no conversations took place inside that projection in the previous test, another round of testing was planned, considering the use of a pre-existing public gobo lamp projecting a static pattern on the pedestrian street [Fig.28]. This time, the goal was to understand whether people would be ready to engage in a conversation with a stranger within a light projection (once a first user had already engaged with the projection). The idea was to place different icons and keywords inside the projection, with a person in the middle waiting for a second user to join, with the help of a sign inviting passerbys for a talk. This presented the opportunity to inquire on the current approaches of

the Municipality in the use of projections in urban design, with an interview about their light design project in Friisgatan. The landscape architects at Malmö Stad responsible for the project mentioned that the renovation of the urban plan of Friisgatan included a light installation because *“a central ‘square’ on the street could strengthen the community*”, while serving the purpose of placemaking, giving *“a sense of place: I am here at Friisgatan again…”*. Elements mentioned in their considerations throughout the design that would be relevant for this project and change the course of this round of testing and the subsequent final design are also:

- *Moving light can be stressful to certain humans and animals*
- *Big thrill design quickly grows old*
- *When there was no gobo pattern in the lamp (only light) the focused light on the square was used for various video recordings of dance and so forth. Like an urban stage. This however led to people living in the area to complain about the noise.*



“*The idea was to give the square a different vibe than the surroundings and give it a quite dramatic change between day and night. The square is deliberately darker than the rest and therefore suitable for projections.”*

Fig. 28: The light installation in Friisgatan used in this testing phase

When asked for permission to conduct the test in this space, the urban architects declined the initial request for “*traffic safety reasons*”, not allowing drawings, texts or other kinds of markings in streets. “*The only markings that are permitted in street scapes are traffic directions or signages*”. This made emerge other challenges connected with testing and designing experiences in public spaces, and changed the settings for the testing. At the end, the designer decided to stand under the projection with a hand-held sign displaying “Wanna talk?” for an hour on a Friday night.

This resulted in four brief conversations with youths (three immigrants and a couple of Swedes). After being introduced to the context of the project, the users would be invited to stand on one of the three surrounding shapes, depending whether they wanted to talk about the city, local food, or Swedish culture. When being asked for their comments after the conversation, one of the people mentioned “*if it weren’t because of this [projection], I would not have stopped to talk with a random stranger*” and that the sign helped clarify the context of my presence, effectively acting as a call to action.

(SEMI)FINAL DESIGN

Sketching, lo-fi prototyping, and testing different elements and variations of the possible interface and experience of the citizen’s proposal for an augmented yapping station led to understanding better opportunities and challenges behind the idea. The result is described here as a semifinal design, as it explores one of the possible developments of Video Stations and Augmented

Squares, while admitting a need for further iterations and testing through a situated and participatory approach for its adaptation to serve specific locations and communities.

The concept aims to create third places through projected ice-breakers around the city, equipping Malmö with strategically-placed augmented streetlamps indicated by illuminated signs and descriptions of its usage. There would be solar-powered projectors secured to a pole imitating a streetlamp (or repurposed lamps) which, triggered by a motion sensor connected to a single-board computer, would project on the square an animation as screensaver and react in the presence of citizens by playing different visuals to support them starting a conversation (with others or themselves). The media and experience presented here focus on proposing keywords, icons, and questions as conversation topics for young immigrants to chat about Malmö landmarks and some Swedish concepts, such as *fika* or *lagom*. By engaging with the projections, citizens have a chance to come together and explore how their experience and background meet the local cultural context, while possibly making a friend. This (semi)final iteration of the concept is hereafter presented with a sketch of the projections [Fig. 29] and the flowchart of the experience [Fig. 30], with the Participant Journey Map (Mast et al., 2021) in mind.

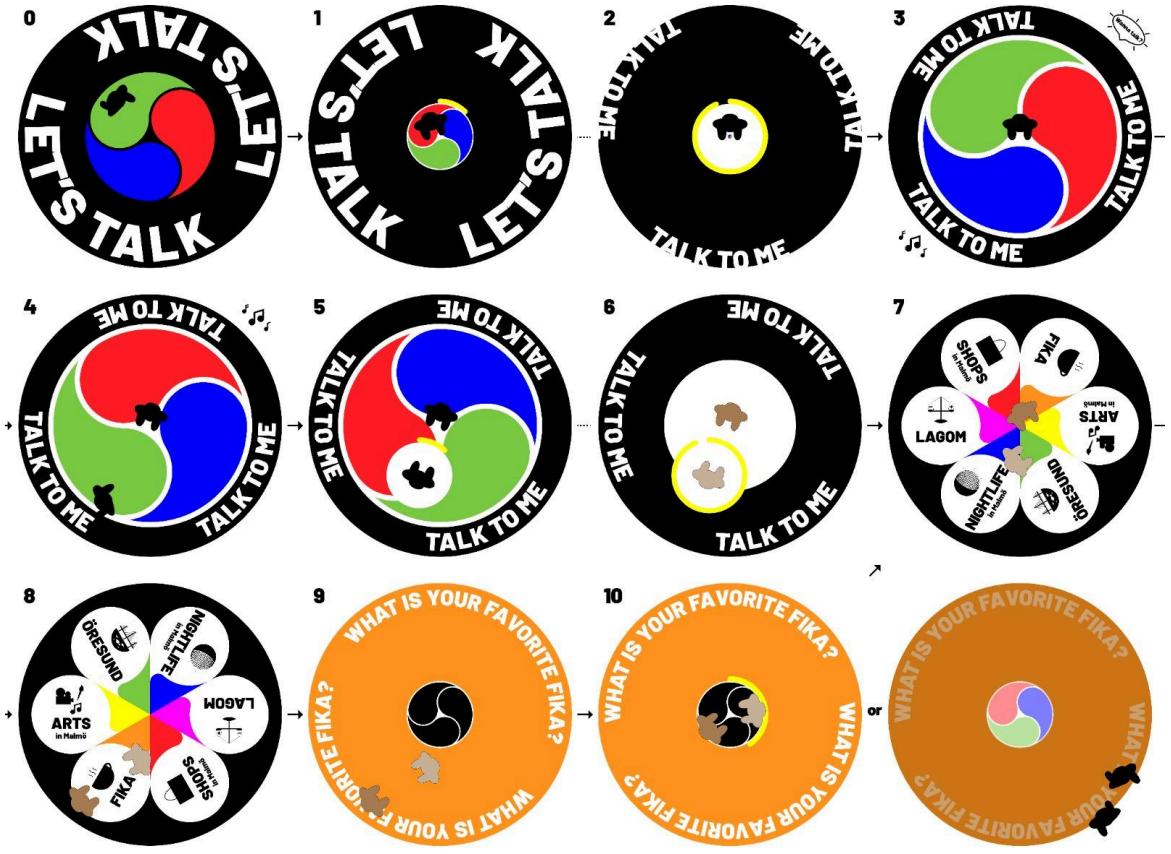


Fig. 29: Sketch of the storyboard of the animations for the final prototype (aerial view of the experience)

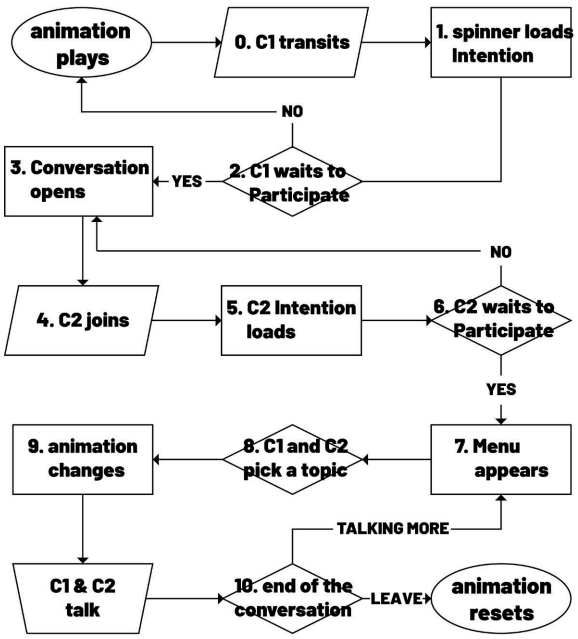


Fig. 30: Flowchart of the experience

The experience would consist of:

0. Player 1 - or rather Citizen 1 (C1) walks through a slow animation of the dormant interface (as in a screensaver) that seeks to lead them from Transit to Awareness.

1. If they stop for over 5 seconds, a sensor activates a spinner loading around their mapped figure, while the "screensaver" contracts into the center. Raise Interest and register Intention.

2. A pulsating white circle appears representing the personal space of C1, and the rotating written message changes as the spinner finishes loading, marking the start of Participation.

3. The rotating icon reappears with a different Call to Action, starting the conversation. At this time, a vertical sign "Wanna Chat?" lights up near the streetlamp and a warm vibrant sound plays what could be compared to a quick jingle or a ringtone.

4. The centripetal animation plays seeking to invite others into its centre, with the help of other low-impact audiovisual cues, such as the illuminated sign and an ambient sound.

5. When a Citizen 2 (C2) and eventually more (C2, C3, etc.) stop prolongedly while crossing through the projection, a spinner appears around their personal space.

6. As it finishes loading, the animation changes, unveiling the personal space of C1, now expanded and connecting with C2.

7. An animation fans in a "menu" of conversational topics connected to the city. Each element should come with an icon or a pictogram.

8. By moving together on one of the topics, C1 and C2 can select an option and start loading a spinner.

9. The animation changes to open the selected option, proposing one random question from a list of kick-off questions. The background may also include the use of photography and videos in connection to the text.

10. C1 and C2 may open the menu (7) again by standing together in the centre of the projection, on top of the main icon, in stand-by.

In case they want to leave (anytime during the experience), they can move away from the projection, where the main icon would expand back to 0. If one person leaves the experience but the other stays prolongedly, the experience goes back to 1.

Additionally:

If C1 and C2 (or more people) approach together, the experience would start directly at 6. Everyone's personal space would be mapped, to merge together into a large spinner.

If no one joins C1 within a few minutes, the program leads C1 to 7 in a solo journey, resulting in an introspective experience.

A button under the streetlamp should grant the accessibility of the experience by providing guidance and an alternative audio interface for people with reduced sight.

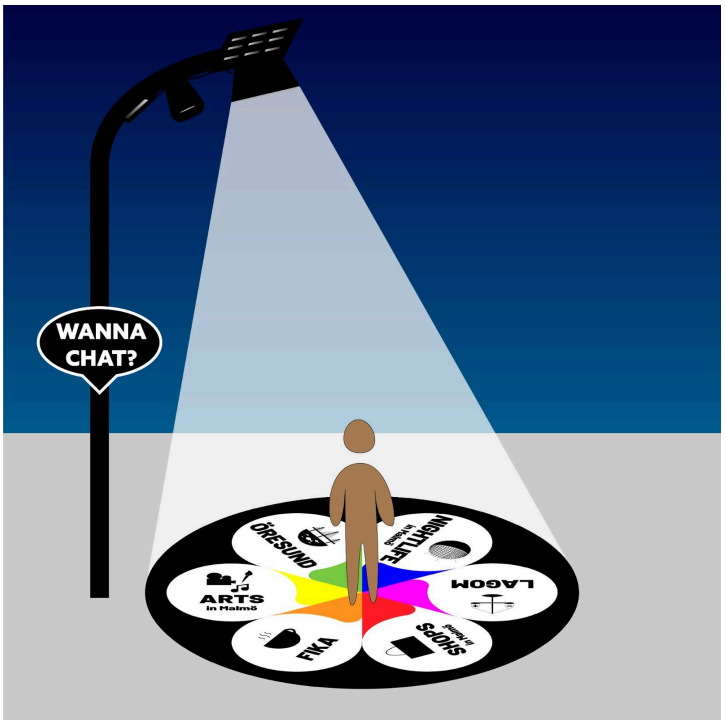


Fig. 31: A digital sketch of the streetlamp

Nonetheless, this is only one Video Station of a possible service system, and it invites for its replication and adaptation with different experiences, to tackle the diversity among the citizens of Malmö. A networked platform of different projectors distributed around the city [Fig. 32] could replicate a geocaching experience, where each streetlamp can offer an experience representative of the community, and a reason to visit a new area of Malmö.

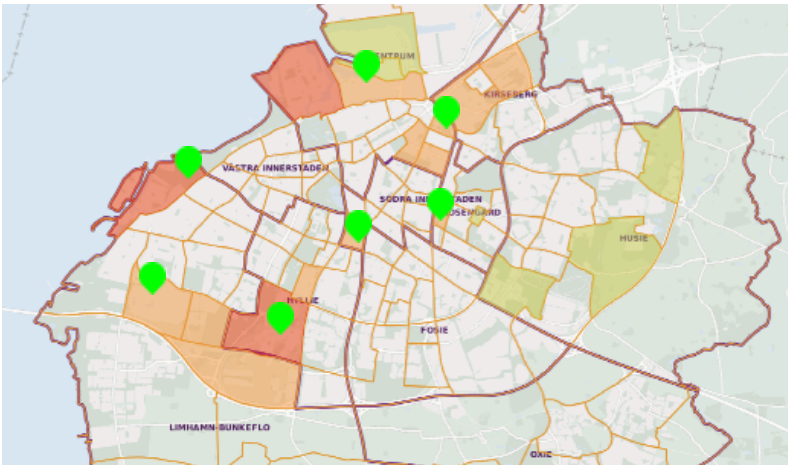


Fig. 32: Possible distribution of locations for a network of augmented streetlamps

RESULTS

NEEDS AND OPPORTUNITIES

The research phase highlighted some of the challenges young immigrants aged 20-34 face trying to integrate into the urban and socio-cultural fabric of Malmö. This is a major demographic group in the city (Malmö Stad, 2014) so the project reached out to a sample group of ten citizens with different cultural backgrounds to interview them about their expression in and impressions of Malmö. They shared a need for offline community-making and free cultural activities (especially outdoors and during winter), difficulties engaging with local Swedish people, struggles with the language barrier, and a lack of acquaintance with less central areas of the city.

In connection to the predictions for the population distribution in Malmö, some hotspots to be possibly activated through urban and cultural regeneration were also identified [Fig. 14 & 32].

CITIZEN-CENTRIC DESIGN

Focusing on this target in relation to the city served as an opportunity to explore a citizen-centric design approach, exploring a vision of Urban Interaction Design in line with some understandings of smart cities. Adaptations to common design methods (interviews, personas, and workshops) allowed a focus on citizens and their urban interactions. This supported studying citizenship, not only as civic engagement, but as the relationship between the citizens, communities, places, technologies, and services of the city (also in a perspective of urban community-making), and as an anchoring element for the participatory process, where citizens are knowledge producers, co-ideators, and co-designers of their own urban futures. In this process, the workshop included a focus group on sharing spaces, a sketching session, and a conversation for collaborative forecasting.

FACE-TO-FACE DIALOGUE AS A DESIGN METHOD AND SOLUTION

Throughout the process, conversations emerged as a method initially unaccounted for. Dialogue with the citizens and among them demonstrated to be central as a co-design approach, as well as a core dynamic and goal in the final design concept.

The workshop remarked how face-to-face settings are beneficial and preferred for social interaction, and that no interface or device should replace the warmth of real-life human connection. This acknowledgement led the citizen co-designers to focus on a “yapping station” as the preferred evolution of the Augmented Social Streetlamps resulting from the use of AR in public spaces.

PUBLIC AUGMENTED REALITY

The collaborative process exploring possible opportunities around augmented public squares and third spaces resulted in the (semi)final concept of Augmented Social Streetlamps, in response to the research question *How may projections augment public spaces to facilitate cross-cultural dialogue and mediate social inclusion and cultural belonging?* The initial thesis here investigated led to the hypothesis that Augmented Squares and Video Stations projecting icebreakers in the city may support the local experience of young immigrant citizens. This is sketched, prototyped and tested

upon with the targeted citizens, to assess, improve and iterate on the concept. Signs of engagement and positive sentiment were observed among the testers, but further research and co-development is needed. Nonetheless, the project suggested that XR can assist social public life in urban spaces.

Testing showed that projections can influence spatial perception and even intrigue citizens and nudge them into getting involved, potentially resulting in placemaking tools that invite citizens and host third places, supporting urban community-making in the long term. However, an activation button may limit users from getting from Transit, Awareness and Interest through to Intention and Participation. Instead, clear verbal calls to action and icons/pictograms had a positive impact.

Projecting on the floor allows designers to treat the square as a collaborative screen, where the users break free from their formal and passive role of frontal audience (e.g. such as in a cinema) to room around, face each other and socialise. If mapped, the movement of their body can then become their medium to control the interface, becoming an embodied cursor.

DISCUSSION

This thesis aimed at exploring the potential of device-free Extended Reality for inclusive smart cities, with a focus on the population of Malmö. Through an iterative reflective process of design research (Gould & Lewis, 1985), the designer facilitated the bottom-up proposal of an urban platform of augmented social streetlamps to create digiphysical third places in public spaces and support immigrant citizens aged 25-34 meeting other people to discuss about the city and Swedish culture. As a start, different parts of the prototype showed engagement, but the impact of the fully working prototype would require a long-term collaboration with local institutions and financial support for its technical development. Nonetheless, the project represents an exploration of a citizen-centric approach to Human-Computer Interaction, Human-Media Interaction, and Urban Interaction Design, and a glimpse into possible urban and social applications of digital projections, Spatial Augmented Reality and Augmented Play Spaces.

AUGMENTED SOCIAL STREETLAMPS

TAKEAWAYS AND OBSERVATIONS

While the project does not intend to offer a solution to a problem, it wants to give an empirical contribution to the design research of possible future augmented public spaces, with a tentative design concept with, by, and for the citizens of Malmö to be reappropriated and further developed. This design proposal seeks ways to encourage sociability and tackle loneliness, prototyping and testing one approach suggested by the citizen co-designers: questions as icebreakers and conversation starters in public spaces.

This was the result of a process participated by its target, which together with the design facilitator focused on social sustainability by concentrating on their need for community and cultural activities, to better understand Swedish culture and to reconnect with the different areas of the city. By prioritising immigrant young adults while keeping in mind principles of Universal Design (Connell et al., 1997), the proposal wants to be tailored to some, but accessible to all (or to the most - more

realistically). However, the project involved only a small group of participants partially representative of the local census, so further refinement of the project would benefit from a bigger number of participants. Similarly, the role of local Swedish populations (especially the ones located where the installations would take place) should be defined, as well as that of indigenous knowledge and more-than-human actors.

How may urban XR and projection mapping create cultural inclusion and belonging? The citizens participating in co-design found that they may do so by incentivising dialogical experiences. While this may have been biased by some elements in the structure of the workshop (such as the initial dialogical activity, or the projections on the floor), the testing showed promising results that should however be further verified. For now, icebreakers were proposed to be projected on the ground in public spaces, and iterative prototyping and testing led to elaborate a proposal of their experience and interface. These are some of the key findings:

- Young adults live a conflict between their reliance on technology and their need for offline spaces. This may be faced through device-free projections that do not separate them with screens, but bring them together face-to-face.
- Citizen's approach to public space is rarely stationary for prolonged times. Also, youths in the square tend to stand, especially if they are there for a short time. Even if sitting seems comforting in more private spaces, it is not observed to be their preferred choice outdoors. While this results only from limited observations and may be biased, also because there are already social benches in the area -"Pratarbänk"- this project would not design for sitting.
- The failure of an activating button led to the understanding that the projections would need to be always active (within set times) and reactive, to overcome the challenge of getting the users to activate the media.
- Texts and icons showed to be useful in communicating the experience to the users, so a visographic approach is recommended: explicit verbal suggestions (such as keywords or kick-off questions) are to be complemented by non-verbal cues, such as icons/isotypes.
- Some people seem to benefit from an incentive supporting them in new social interactions (a sign with a verbal cue, in this case). However, the time and location of the test may have had an impact on the results, so it is to be further tested outdoors in the evening/night.
- The first interaction with the experience seems to be more challenging than a latter interaction with a stranger in the square that already started it (but variables such as gender, personality, culture and other environmental factors may play an important role). This calls for more testing, with a focus on the transition from Awareness to Intention and Participation.
- Light design and visual projections can alter the perception of space, so light may be treated as part of the urban furniture (not only as a service - lighting).
- Saturated visuals and fast animations may have negative impacts on traffic safety, and more-than-humans, so slow animations and natural elements may be recommended.
- Place-specific prototyping and testing in public spaces requires the collaboration of public institutions, permits to use public soil and additional bureaucratic work, other than the financial backing for the technical equipment, the scouting of set locations, the involvement of situated communities, the mapping of human and more-than-human actors and stakeholders, and the consequential development and contextualisation of the experience, so it may represent an important challenge in a short-term project such as this.

Difficulties during the different testing sessions also provided useful observations that may lead to recommendations when engaging with projects for public urban spaces:

- The first test was conducted indoors in the (semi)public facilities of the university in the hope that no formal permissions would be needed, there would be many young immigrants (the target) passing through, and there would be access to electricity (for the initial intended use of a projector). While part of these assumptions were wrong (as the use of the projector was not allowed), the presence of the target supported gathering some data: as debriefed above, the button did not work, but worded calls to action helped connect strangers into a conversation.
- The second and third test happened outdoors and in the evening, confirming the potential of light (together with words and icons) in placemaking and starting/feeding conversations. The integration of a sign proved to be positive, but a need for context and instructions surfaced. However, it is to be noted that the close presence of the designer in both these tests may have generated biases in the data.
- Additional rounds of testing would have been beneficial to verify how the (semi)final experience and interface (artefact and animations) resulting from previous iterations would be welcomed and explored in urban spaces. The lack of time, resources, and formal access to public soil did not allow this to be concluded within this project.

FUTURE CHALLENGES

Getting C1 to stop in a public space and wait alone under the spotlight of the projector may remain the main challenge. However, once the infrastructure is available and trust is established, its access would possibly become more familiar and common through time: the collaboration with institutional stakeholders could give the installation visibility and credibility and support the integration and use of the technology. Long-term testing would be needed.

Adapting the experience not only to bind strangers, but also to be enjoyed with a familiar group of friends, or possibly even alone might be required. Consequently, the number of users to be possibly hosted by the experience is to be better understood.

Further testing would require appropriate resources and equipment, such as an outdoor projector (with appropriate brightness), a visual programming or projection mapping software, a motion tracker, and eventually urban furniture or props to map and project on, to also explore approaches to Mixed Reality.

Developing culturally aware and inclusive aesthetics is another ambitious challenge that must be put to the follow-up of the project, as a superficial approach to the matter risks resulting in either a redundant or tokenistic result. For better representation, the project could be open for citizens and communities to develop their own media, integrating the urban platform with product reciprocity, where “one person's product is another person's platform” (Brynskov & al., 2014: 58). This applies also in the case of designing a multisensorial system to involve others. Visual, sound, and tactile experiences are some of the elements that could contribute to an effective and immersive projected augmentation, but would require more research and development: light-reactive prints, relief patterns, biomimicry in visuals, and even additional artefacts, furniture, or kinetic elements. Depending on the projection surfaces, the content may be 2D or 3D. Multilingual access to media and interface could also be beneficial.

The place-specific design, as well as the distribution of the experience around the city, also requires more time and resources. Outdoors testing and wild prototyping could then be useful. A

variety of experiences could be tested and developed, also for a more citizen-centric and inclusive result. The solution here tested and presented is to be considered just a first iteration of the concept.

Future iterations may reconsider some of the ideas from the ideation and sketching discussed here, focusing on their main interactive properties and the role of projections in the experience. While these are here gathered around their main features, they are not mutually exclusive: In augmented sport courts, board games, or role-play, projection may enhance playfulness, sensing the position of the players and elements of the games within a set environment, framework, and rules, possibly tracking points or progress. “The floor is lava” was repeatedly mentioned by the citizens involved. Digital gardens, community rooms, bulletin boards, and drawing boards would work through (temporary) memory, gathering and screening traces from the citizens engaging with the artefact. An artistic approach could be developed with a dancehall, stage, or reader, where projections would have an assistive role, as guidance through a performative activity. Similarly, educational experiences could range from a salsa lesson to a step-by-step repair station.

The content of projections could also be in the hands of citizens (or community representatives), who may propose media, or stream an activity, with a remote workshop station for skill-sharing and group outdoor attendance of activities happening at another video station. Finally, augmented streetlamps may even simply set an ambience, whether for an active activity, as in a Zumba class or dancehall, or some more relaxing, as mindfulness, and even prayer. An opportunity for “branded” content for city events and holidays may also be further explored.

Augmented Streetlamps and Augmented Public Squares may represent a multiplicity of chances for smart citizenship, even promoting transparency in the management of the city and sharing decision-making with citizens, developing different wayfinding solutions, promoting experiences for citizen science, and more.

CONCERNS

While the technical safety of the infrastructure should be assessed before making the experience public, other elements are to be considered. For instance, the dependence of projections on darkness should raise concerns around the feeling of safety of its users, so the location of the artefact may be of particular relevance for its usability. Parallelly, lighting, media, and aesthetics of the streetlamps may be further designed to make people feel safer. However, it may be more appealing to groups, rather than to individuals at night. Interacting with the nightlife of the city, its users may also not always be sober, and this could be studied in further iterations.

Interactions will make temporary use of citizens’ data, as through motion sensors, to bring an activity to an end (e.g. as for their position in spaces or tracking points in games). While no sensible information seems to be needed at this stage of the project, and no data should be exposed outside the time of the experience itself, if that changes in future iterations, citizens should be protected from possible breaches of their data. Ideally, for ethical reasons, no facial recognition should be used, and no personal data should be stored (unless for the technical improvement of the experience). With the expansion of mass-survaillance technologies, this project should not be weaponised to further the control on citizens’ data, using it against them rather than in their favor.

It is also worth mentioning that even when using device-free AR to acknowledge and tackle the digital divide, tech risks remain partly divisive and unequal, as previous access and familiarity with similar technologies, as well as their media literacy, could impact whether and how different citizens approach the experience.

The reliance on Media Architecture to promote social sustainability also comes with risks. As Dourish & Bell (2011) warn, IoT may enact “an explicit act of colonization” aimed at the sales of ICTs until saturation. With the possible commercialisation of urban augmented spaces, these, instead of fighting technofeudalism, may end up feeding it.

Lastly, gentrification is another possible side effect of a successful design that activates peripheral urban areas, but it is finally up to the institutions to do what possible to grant access to housing and competitive commerce.

AUGMENTED SQUARES

The project reinforced the conviction that SAR and APS have placemaking potential, and through light, media and eventually play, can really facilitate the interaction between citizens and with the city. These cannot however be designed without them. As public spaces and the data they recollect belong to the people, they should be involved in the design of its services, putting their needs on the forefront, for cities to truly allow them to express their citizen-ability (Foth, 2018). The local legibility, literacy, and legitimacy cities carry (Dourish & Bell, 2011) become a reminder of how the urban space ought to be representative and descriptive of the reality, living, and culture of its citizens, and even of their joys and sorrows. To succeed, citizens' participation in UIxD is essential and a collaborative and citizen-centric approach to urban informatics has to be devised. Only then - and once granted democratically public access to tech - can cities aim to tackle the digital barriers (Soares de Oliveira Neto & Kofuji, 2016), the digital divide (van Dijk, 2006), loneliness and radicalisation.

And if the city belongs to the citizens, it is fair to ask: should they not also own the media production? Within a broader scope, the project should aim at giving citizens more agency and even ownership over the media, for a more inclusive and citizen-centric result, becoming an opportunity for citizen development and for people and communities to design these experiences together, both as permanent installations or temporary “events”, supporting them with customisable augmented spaces to provide a digital audiovisual layer to their activities (e.g. as a stage for a choir, a dancefloor for training, or a board for public discussions). A variety of ideas and concepts were drafted in the process but omitted in this paper – from a projected community permagarden growing through user interactions to a service allowing different temporary augmented customisations of public space – and all point at the underexplored value of augmented public spaces, squares, and playgrounds. With an application comparable to that of screens and portals, projecting on a vertical wall, a corner, or a holographic display can also allow a public streaming service for similar use, allowing the projection of workshops, skill-sharing, and community activities in multiple public spaces of the city at the same time, with the objective of reducing the distances and extending the participation.

The extent of what are the possible installations, experiences, and media to be designed is difficult to measure and only situated communities can really say what should be further designed and tested. However, within its limited scope, this project aims primarily to contribute to the growing

interest in using Extended Reality, Media Architecture, and projection mapping for active citizenship and community-making, where citizens are at the centre of the design process and even the owners of the media. To translate this into a project of urban commons and as a multistakeholder service that can benefit all, how citizens can create media collaboratively (regardless of their digital literacy), should be further explored.

CITIZEN-CENTRISM

A citizen-centric approach is already recommended from policy making to digital governance, and it calls for further research in Urban Interaction Design. Some of its understandings in the context of cities and urban services invite for responsiveness to citizens' needs, transparency of their data and information, and inclusion of the citizens in decision-making, whether through their data, in co-design, or/and with monitoring and feedback (Kamaruddin & Noor, 2013; Cardullo & Kitchin, 2018; Lee et al., 2020; Dechamp et al., 2025).

Openings for a citizen-centric approach also to XR, SAR and APS emerged, with a need to document best practice and study long-term success stories across the globe (also outside North-Western societies). However, it is not up to this thesis to dictate whether the right direction would lead through a definition of "citizen-centric design", its requisites or some standardised processes or framework, as it aims to be solely an invitation to reflect on the matter.

The project tried to apply it with a focus on citizens in research, ideation, and testing. Starting with citizen interviews (with participants from different districts of Malmö, origins and cultures, jobs, and hobbies) to then involve them in co-ideation, it put citizens and their needs at the lead of the design process while focusing on the relationship between citizens, places, and technologies.

The intention to further discuss a citizen-centric approach here instantiated represents just an observation for a transdisciplinary discussion and exchange of best practices, rather than the pretence of structuring a framework for a methodology. This would require not only a comparative study focusing on the inhabitants of different urban spaces, but also an extended and iterative exploration of their relationship with the city and citizenship. This may be explored through (auto)ethnographic tools and cultural probes, street interviews, joint city walks, workshops at the park, and other activities taking place in the city, always with an eye on the intersectional realities behind the deeply diverse communities partaking in city-making.

This project focuses on how simple considerations and adaptations may help set the focus on smart citizenship. For an inclusive approach, during the process, a definition of intersectional citizen is taken into account, to promote a universal approach that considers different needs and marginal minorities: it sets some of the elements of the identity and ability of citizens that may influence the design result. Inclusive universal design may require designing always for a radical target, keeping in mind ability, ethnicity, gender, culture, religion, and the users' marginality in all, unless stated otherwise. In this case, the target is set to be 25-34, recently migrated to Malmö, and originally from different cultures, to not focus only on the ruling cultural hegemonies and challenge Eurocentrism in design and urban planning.

A limitation of the technological approach adopted here is that the experience may be mostly accessible to people with sight, and the choice of content/media would currently target people able to read English. This leaves space for further improvement of the concept and invites exploring other possible media experiences to be hosted by the interface.

CONCLUSIONS

How can public uses of projection mapping activate urban spaces and bring people together? Through reactive low-impact animations of textual, visual, and multisensorial media, device-free Extended Reality may create third places and support cross-cultural dialogue for inclusive smart cities. In this citizen-centric design research process, interviews led to acknowledging the struggles of young immigrants connecting with the people and the city of Malmö, to then involve them in a co-design workshop exploring sharing spaces and projection settings. Empowered citizens took over the process and proposed a system of Social Augmented Streetlamps to provide icebreakers in urban areas such as squares and nudge citizens to socialise -e.g through proposed conversation topics. Sketching and prototyping the potential artefact and service, with its components and affordances, led to testing, making some of the challenges and opportunities emerge, gathering feedback for future iterations and confirming projections as of possible support for social sustainability.

This short project aims to contribute to the field of HCI, HMI, and Urban Interaction Design by exploring public social uses for Spatial Augmented Reality through a citizen-centric approach to media architecture and platform urbanism, resulting in an invitation to create urban commons for a public augmented urban reality and digiphysical third places, where citizens are designers, producers, and consumers. The (semi)final design opens the discussion on Augmented Squares and Video Stations, calling for further research on ways to democratise the access to XR and center citizens in the design of urban informatics for them to seize the media production.

ETHICAL CONSIDERATIONS

The implementation of a citizen-centric approach in this thesis focuses on a specific range of the population, leaving others out of the design process, at this phase of the project, with the risk of biases and misrepresentation. With more resources and long-term collaborations with public institutions to scale up the project, more co-designers may be reached.

Universalisms when designing for culture may be challenging. An inclusive and accessible approach in design should be imperative, but the idea of a common global vision of shared aesthetics and symbolisms is a dangerous (dis)illusion. Finding "the right" audiovisual code to tune the project in for it to create an inviting shared safe space for dialogue and exchange for each one of the over 180 nationalities Malmö hosts would be as ambitious as pretentious. However, in the eventual continuation of the project, a suitable visual style is worth researching. The further exploration of a citizen-centric, culturally-aware, bio-inclusive, and indigenous-acknowledging approach to spatial design and multimedia would require extended ethnographic research, participatory methods, and local engagement.

An approach to Indigenous local culture and (eventually lost) traditions can also pave the way for ancestral futures. In this case, the oral practice is the protagonist, but further inclusion of and collaboration with Indigenous knowledge may add value to such a project.

More-than-human species should also be carefully taken into consideration in the eventual follow-up of this design process, as critters may also have important parts in the urban ecosystem.

Finally, smart city projects cannot disregard concerns around environmental sustainability, considering the use of renewable energies and green cloud computing, and the assessment of the general environmental impact of an experience. These would also become necessary for the realisation of this project.

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